



A new bivariate Poisson distribution via conditional specification: properties and applications

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ABSTRACT

In this article, we discuss a bivariate Poisson distribution whose conditionals are univariate Poisson distributions and the marginals are not Poisson which exhibits negative correlation. Some useful structural properties of this distribution namely marginals, moments, generating functions, stochastic ordering are investigated. Simple proofs of negative correlation, marginal over-dispersion, distribution of sum and conditional given the sum are also derived. The distribution is shown to be a member of the multi-parameter exponential family and some natural but useful consequences are also outlined. Parameter estimation with maximum likelihood is implemented. Copula-based simulation experiments are carried out using Bivariate Normal and the Farlie–Gumbel–Morgenstern copulas to assess how the model behaves in dealing with the situation. Finally, the distribution is fitted to seven bivariate count data sets with an inherent negative correlation to illustrate suitability.

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1. Introduction

Bivariate discrete Poisson distributions have enjoyed a good amount of attention over the last couple of decades or so. These distributions are quite useful in modeling paired count data exhibiting correlation and possibly representing over and under dispersion. Paired count data arise in a wide context including marketing (number of purchases of different products), epidemiology (incidents of different diseases in a series of districts), accident analysis (number of road accidents in a site before and after infrastructure changes), medical research (the number of seizures before and after treatment), sports (the number of goals scored by each one of the two opponent teams in soccer), econometrics (number of voluntary and involuntary job changes), survival analysis, queuing and branching process just to name a few. However, not much is discussed and studied in thorough details on a correlated bivariate Poisson distribution with a negative correlation. The most celebrated form of a bivariate Poisson distribution that starts with three independent Poisson random variables X_i , $i = 1, 2, 3$ with $X_i \sim \text{Poisson}(\lambda_i)$. Then, the random variables $Y_1 = X_1 + X_3$, and $Y_2 = X_2 + X_3$ is said to follow jointly a bivariate Poisson distribution with the following

probability mass function (p.m.f.)

$$P(Y_1 = y_1, Y_2 = y_2) = \exp(-[\lambda_1 + \lambda_2 + \lambda_3]) \frac{\lambda_1^{y_1} \lambda_2^{y_2}}{y_1! y_2!} \sum_{i=0}^{\min\{y_1, y_2\}} \binom{y_1}{i} \binom{y_2}{i} i \left[\frac{\lambda_3}{\lambda_1 + \lambda_2} \right]^i.$$

For a comprehensive treatment of the bivariate Poisson distribution and its multivariate extensions, the reader can refer to [12] and [9]. Paul and Ho [21] discussed in the estimation of the bivariate Poisson distribution and hypothesis testing regarding independence.

The major limitation of the above bivariate distribution is that it is applicable to data which allows for positive dependence between the two random variables only. Additionally, since its marginal distributions are Poisson (with parameters $\lambda_1 + \lambda_3$ and $\lambda_2 + \lambda_3$ respectively), it cannot be used to model over-dispersion/under-dispersion. Several earlier works on the bivariate Poisson distribution utilizes the above joint p.m.f. Mimicking the words/phrases provided in [14], ‘It should be worth noting that the bivariate Poisson distribution reported by Teicher [24], Campbell [5], Holgate [10] has the inherent limitation that the correlation is necessarily positive, and hence is not useful in modeling the situations (e.g. in biology) involving competition for limited resources where a negative correlation is expected.’ As a consequence, Lakshminarayana et al. [14] studied a bivariate distribution whose marginals are Poisson developed as a product of Poisson marginals with a multiplicative factor. The resulting model exhibits both positive and negative correlation. The one apparent limitation of the bivariate Poisson model as described in [14] could be that in the case of a boundary value of the dependence parameter (i.e. λ_3 assuming a very small negative value), the model might fail to adequately fit the data and may provide a false positive scenario. Subsequently, any test for independence in such a case under the classical likelihood ratio set up might not be appropriate.

Motivated by this, we consider a different construction approach, based on conditional specification (for details, see [3] and the references cited therein) to construct a bivariate Poisson distribution. In this approach, we start with two conditionals which are Poisson type with the parameter(s) depending on the conditioning variable and obtain the most general form of a bivariate discrete Poisson distribution that accounts for negative correlation. The associated marginal distributions of X and Y can cater to both over and under dispersion under certain parametric restrictions as shown in Section 2. Noticeably, the marginals are no more Poisson. Therefore, the resulting model can be applied to model over-dispersion/under-dispersion as the case may be. Moreover, another motivation is that we can not have Poisson conditionals with positive correlation. Therefore, in a real-life scenario, if we have dependent variables whose random nature can well be approximated by a Poisson distribution, we can be sure of the fact that the inherent dependence structure will be negative. As a consequence, the bivariate Poisson distribution via conditional specification which is considered here will be the most preferred choice. This is a salient feature which outperforms all such bivariate Poisson models discussed and developed earlier in the literature. In the literature, several truncated versions of the bivariate Poisson distribution are also considered. Among them, noteworthy of mention is a work by Hamdan [8], who introduced and studied a truncated bivariate Poisson distribution with missing zeros and discussed the method of moments for a particular scenario of parametric restriction.

The p.m.f. of the truncated bivariate Poisson distribution with missing zeros is given by

$$P_1(Y_1 = y_1, Y_2 = y_2) = V^{-1} \exp(-[a + b + d]) \sum_{i=0}^{\min\{y_1, y_2\}} \frac{d^{x_3} a^{y_1 - x_3} d^{x_3} b^{y_2 - x_3}}{x_3(y - x_3)},$$

where $V = 1 - \exp(a + d) - \exp(b + d) - \exp(a + b + d)$, $(y_1, y_2) \in C$, where $C = \{Y_1 \geq 1, Y_2 \geq 1\}$.

Adamidis and Loukas [1] discussed the estimation of the above model using the EM algorithm. Dahiya [6] discussed the method of maximum-likelihood estimates for the special case $a = b$. Kocherlakota and Kocherlakota [13] studied some problems associated with count data from a bivariate Poisson distribution, in which the marginal means are functions of explanatory variables.

Success of any new model heavily depends on its suitability in handling data observed in real situations. As such after investigating various useful properties, we consider a number negatively correlated bivariate count data sets from different contexts to justify its applicability. In what follows, we will first motivate with five data sets from the highly popular English Premier Football League (<https://datahub.io/sports-data/english-premier-league-data>) where the goal scored by the home team and away team in each game is seen to exhibit negative dependency. We consider data sets arising from five seasons, namely, 2014/15, 2015/16, 2016/17, 2017/18 and 2018/19 in order to understand the general tendency of the negative correlation that exists. For each season we have 380 observed pairs of counts, (Full Time Home Team Goal, Full time Away Team Goal). Both counts (in each data set) are over dispersed and are negatively correlated which are the two chief characteristics of our model. From the findings of our modeling, it can be argued that the proposed distribution provides an adequate fit to four out of these five observed bivariate data sets barring the season 2017/18.

Next, in order to further check the proposed model with some existing three parameter bivariate count models, we consider the example from [2] where the authors have analyzed a negatively correlated bivariate count data that originate as counts of surface faults (X) and interior faults (Y) of lenses. Findings from data fitting reveal that the proposed model is better than McKendrick–Wicksell bivariate Poisson distribution [2], BPHD (the Bivariate Poisson distribution proposed by Holgate [8] and the bivariate geometric [13].

Finally, we consider another negatively correlated bivariate data about the number of seeds and plants grown over a plot of size five square feet and plants grown studied in [20] and [12]. Here the observed small negative correlation is again -0.094 and the proposed distribution fit the data very well.

The rest of the paper is organized as follows. In Section 2, introduce some structural properties of the proposed bivariate Poisson distribution via conditional specification which was described in [3] and provide the expression for associated marginal p.m.f.s, of X and Y. In Section 3, we provide some useful structural properties of the bivariate Poisson distribution described in Section 2. Section 4 deals with the statistical inference of the bivariate Poisson distribution under the classical setup using copula-based simulation study. Some well-known paired count data exhibiting negative correlation are re-analyzed to illustrate the applicability of this distribution in Section 5. Finally, some concluding remarks are provided in Section 6.

2. Bivariate dependent Poisson distribution via conditional specification

Let us assume the following:

- $X|Y = y \sim \text{Poisson}(\lambda_1 \lambda_3^y)$, for each fixed $Y = y$.
- $Y|X = x \sim \text{Poisson}(\lambda_2 \lambda_3^x)$, for each fixed $X = x$.

Here, $(\lambda_1, \lambda_2) > 0$, $0 < \lambda_3 \leq 1$. Note that if $\lambda_3 = 1$, X and Y are independent. According to [3, see Theorem 4.1, page 76], the associated joint p.m.f. will be

$$P(X = x, Y = y) = K(\lambda_1, \lambda_2, \lambda_3) \times \frac{\lambda_1^x \lambda_2^y \lambda_3^{xy}}{x! y!}, \quad (1)$$

where $x = 0, 1, 2, \dots$; $y = 0, 1, 2, \dots$, and $K(\lambda_1, \lambda_2, \lambda_3)$ is the normalizing constant and

$$K^{-1} = K^{-1}(\lambda_1, \lambda_2, \lambda_3) = \sum_{y=0}^{\infty} \frac{\lambda_2^y}{y!} \exp(\lambda_1 \lambda_3^y) = \sum_{x=0}^{\infty} \frac{\lambda_1^x}{x!} \exp(\lambda_2 \lambda_3^x).$$

We will denote (henceforth, in short) the bivariate Poisson distribution of the pair (X, Y) with the p.m.f. in (1) as $BPD(\lambda_1, \lambda_2, \lambda_3)$. Note that the joint p.m.f. in (1) is also known as Obrechhoff's distribution [20]. It is also independently derived in [27].

Note that

(1) The marginal p.m.f. of X will be

$$\begin{aligned} P(X = x) &= K(\lambda_1, \lambda_2, \lambda_3) \frac{\lambda_1^x}{x!} \sum_{y=0}^{\infty} \frac{(\lambda_2 \lambda_3^x)^y}{y!} \\ &= K(\lambda_1, \lambda_2, \lambda_3) \frac{\lambda_1^x}{x!} \exp(\lambda_2 \lambda_3^x), \end{aligned} \quad (2)$$

for $x = 0, 1, 2, \dots$

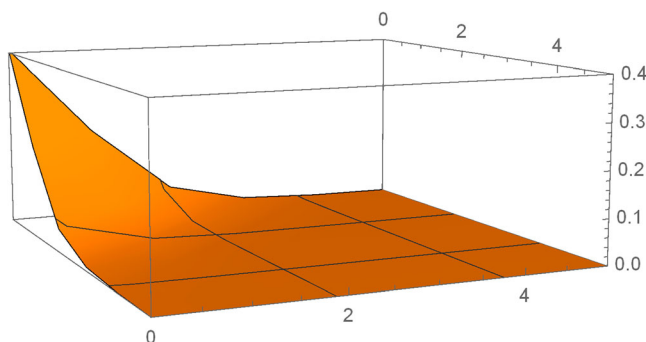


Figure 1. The p.m.f. function for $\lambda_1 = 0.5, \lambda_2 = 0.5, \lambda_3 = 0.5$ with correlation $= -0.2$.

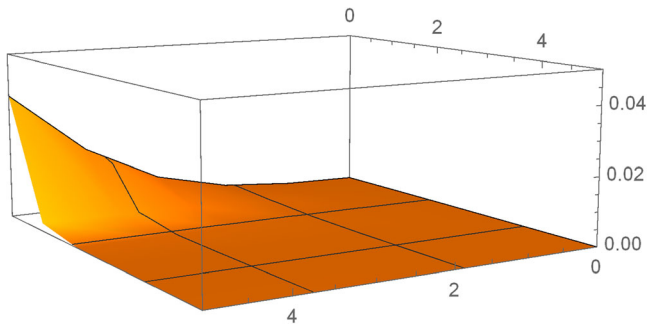


Figure 2. The p.m.f. function for $\lambda_1 = 10, \lambda_2 = 2, \lambda_3 = 0.5$ with correlation $= -0.195$.

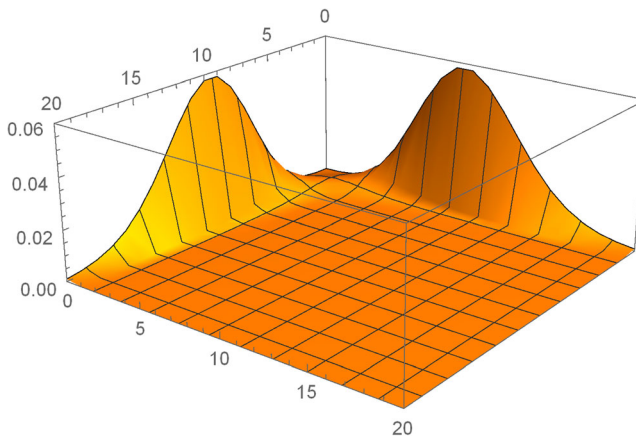


Figure 3. The p.m.f. function for $\lambda_1 = 10, \lambda_2 = 10, \lambda_3 = 0.5$ with correlation $= -0.812$.

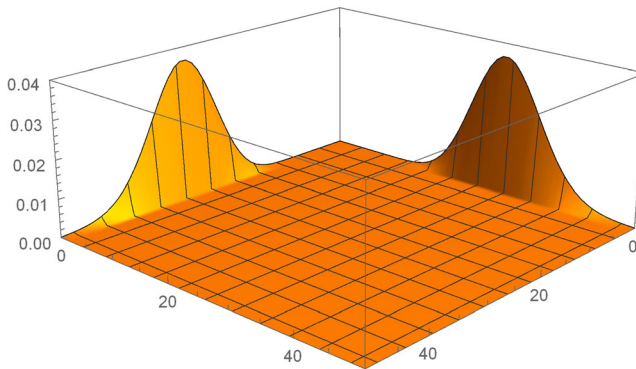


Figure 4. The p.m.f. function for $\lambda_1 = 30, \lambda_2 = 30, \lambda_3 = 0.1$ with correlation $= -0.937$.

(2) Similarly, the marginal p.m.f. of Y will be

$$P(Y = y) = K(\lambda_1, \lambda_2, \lambda_3) \frac{\lambda_2^y}{y!} \exp(\lambda_1 \lambda_3^y), \quad (3)$$

for $y = 0, 1, 2, \dots$

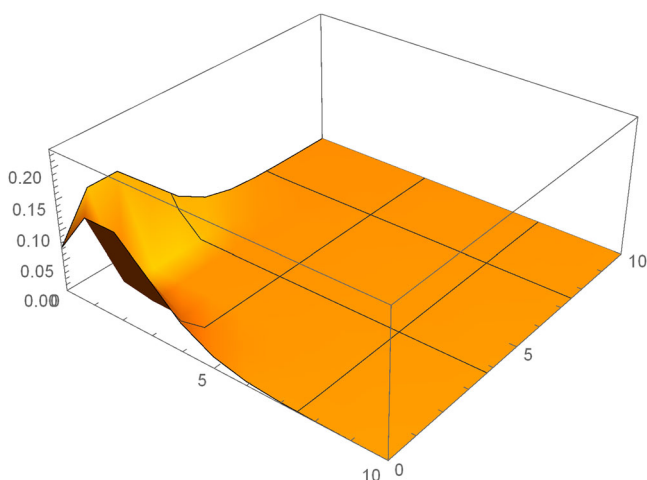


Figure 5. The p.m.f. function for $\lambda_1 = 2, \lambda_2 = 2, \lambda_3 = 0.03$, with correlation $= -0.556$.

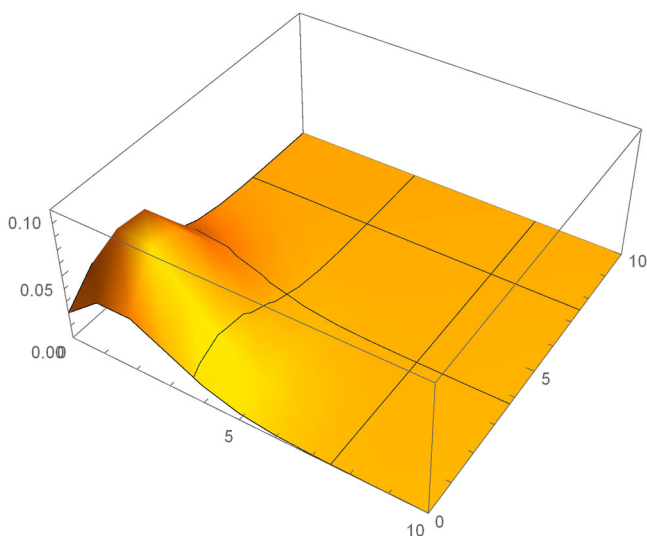


Figure 6. The p.m.f. function for $\lambda_1 = 2, \lambda_2 = 2, \lambda_3 = 0.97$ with correlation $= -0.057$.

- (3) For fixed λ_3 , $P(X = x, Y = y | \lambda_1, \lambda_2, \lambda_3) = P(Y = y, X = x | \lambda_2, \lambda_1, \lambda_3)$. Thus, if $\lambda_1 = \lambda_2 = \lambda$, say then $P(X = x, Y = y | \lambda, \lambda, \lambda_3) = P(Y = y, X = x | \lambda, \lambda, \lambda_3)$. Therefore, in this situation, we get symmetric shapes which are observed in Figures 3 and 4.

Some representative p.m.f. plots for varying parameter choices are provided in Figures 1–6 for the joint p.m.f. in (1). The negative dependence is quite apparent; moreover, it can be easily seen that, for larger values of λ_1 , and λ_2 , the marginals tend to assume a normal shape.

3. Structural properties

Note that since, $X|Y = y \sim \text{Poisson}(\lambda_1 \lambda_3^y)$, $E(X|Y = y) = \lambda_1 \lambda_3^y$. Consequently,

$$\begin{aligned} E(X) &= E_Y(X|Y = y) \\ &= \lambda_1 K(\lambda_1, \lambda_2, \lambda_3) \sum_{y=0}^{\infty} \lambda_3^y \frac{\lambda_2^y}{y!} \exp(\lambda_1 \lambda_3^y) \\ &= \frac{\lambda_1 K(\lambda_1, \lambda_2, \lambda_3)}{K_1}, \end{aligned} \quad (4)$$

where $K_1 = K(\lambda_1, \lambda_2 \lambda_3, \lambda_3)$, and

$$K_1^{-1} = K^{-1}(\lambda_1, \lambda_2 \lambda_3, \lambda_3) = \sum_{y=0}^{\infty} \frac{(\lambda_2 \lambda_3)^y}{y!} \exp(\lambda_1 \lambda_3^y).$$

Similarly,

$$E(Y) = \frac{K(\lambda_1, \lambda_2, \lambda_3) \lambda_2}{K_2}, \quad (5)$$

where

$$K_2^{-1} = K^{-1}(\lambda_1 \lambda_3, \lambda_2, \lambda_3) = \sum_{x=0}^{\infty} \frac{(\lambda_1 \lambda_3)^x}{x!} \exp(\lambda_2 \lambda_3^x).$$

Furthermore,

$$\begin{aligned} E(XY) &= K(\lambda_1, \lambda_2, \lambda_3) \sum_{x=0}^{\infty} \sum_{y=0}^{\infty} \frac{\lambda_1^x \lambda_2^y \lambda_3^{xy}}{(x-1)(y-1)} \\ &= (\lambda_1 \lambda_2 \lambda_3) K(\lambda_1, \lambda_2, \lambda_3) \sum_{\ell_1=0}^{\infty} \sum_{\ell_2=0}^{\infty} \frac{(\lambda_1 \lambda_3)^{\ell_1} (\lambda_2 \lambda_3)^{\ell_2} \lambda_3^{\ell_1 \ell_2}}{\ell_1! \ell_2!} \\ &= \frac{(\lambda_1 \lambda_2 \lambda_3) K(\lambda_1, \lambda_2, \lambda_3)}{K_3}, \end{aligned} \quad (6)$$

where $K_3 = K(\lambda_1 \lambda_3, \lambda_2 \lambda_3, \lambda_3)$, and

$$K_3^{-1} = \sum_{\ell_1=0}^{\infty} \sum_{\ell_2=0}^{\infty} \frac{(\lambda_1 \lambda_3)^{\ell_1} (\lambda_2 \lambda_3)^{\ell_2} \lambda_3^{\ell_1 \ell_2}}{\ell_1! \ell_2!}.$$

One can find the individual variances as well. For example, from the conditional of X given $Y = y$, $\text{Var}(X|Y = y) = \lambda_1 \lambda_3^y$.

Utilizing this fact in the following expression, we can obtain

$$\begin{aligned} \text{Var}(X) &= E(\text{Var}(X|Y=y)) + \text{Var}(E(X|Y=y)) \\ &= \lambda_1 K(\lambda_1, \lambda_2, \lambda_3) \left[\frac{1}{K_1} + \lambda_1 \left\{ \frac{1}{K(\lambda_1, \sqrt{\lambda_2} \lambda_3, \lambda_3)} - \frac{K(\lambda_1, \lambda_2, \lambda_3)}{K_1^2} \right\} \right]. \end{aligned}$$

Therefore, the covariance between X and Y will be

$$\begin{aligned} \text{Cov}(X, Y) &= E(XY) - E(X)E(Y) \\ &= \frac{K(\lambda_1, \lambda_2, \lambda_3)}{K_3} - \left\{ \frac{K(\lambda_1, \lambda_2, \lambda_3) \lambda_1}{K_1} \right\} \left\{ \frac{K(\lambda_1, \lambda_2, \lambda_3) \lambda_2}{K_2} \right\} \\ &= \lambda_1 \lambda_2 K(\lambda_1, \lambda_2, \lambda_3) \left[\frac{\lambda_3}{K_3} - \frac{K(\lambda_1, \lambda_2, \lambda_3)}{K_1 K_2} \right] \\ &= \lambda_1 \lambda_2 K(\lambda_1, \lambda_2, \lambda_3) \left[\frac{\lambda_3}{K(\lambda_1 \lambda_3, \lambda_2 \lambda_3, \lambda_3)} - \frac{K(\lambda_1, \lambda_2, \lambda_3)}{K(\lambda_1 \lambda_3, \lambda_2, \lambda_3) K(\lambda_1, \lambda_2 \lambda_3, \lambda_3)} \right]. \end{aligned} \quad (7)$$

We list some results related to the sum $K^{-1} = K^{-1}(\lambda_1, \lambda_2, \lambda_3)$ which will be used in derivation of results in sequel.

$$\begin{aligned} K^{-1}(0, \lambda_2, \lambda_3) &= \exp(\lambda_2) \\ K^{-1}(\lambda_1, 0, \lambda_3) &= \exp(\lambda_1) \\ K^{-1}(\lambda_1, \lambda_2, 1) &= \exp(\lambda_1 + \lambda_2) \\ K^{-1}(\lambda_1, \lambda_2, 0) &= \exp(\lambda_1 + \lambda_2 - 1) \\ K_2^{-1}(0, 0, \lambda_3) &= K_1^{-1}(0, 0, \lambda_3) = 1 \\ K^{-1}(\lambda_1, \lambda_2, \lambda_3) &= K^{-1}(\lambda_2, \lambda_1, \lambda_3) \\ \frac{d}{d\lambda_1} K^{-1}(\lambda_1, \lambda_2, \lambda_3) &= K^{-1}(\lambda_1, \lambda_2 \lambda_3, \lambda_3) \\ \max\{\exp(\lambda_1), \exp(\lambda_2)\} &< K^{-1}(\lambda_1, \lambda_2, \lambda_3) \\ &< \exp(\lambda_1 + \lambda_2). \end{aligned}$$

Theorem 3.1: If $(X, Y) \sim \text{BPD}(\lambda_1, \lambda_2, \lambda_3)$, then $\text{Cov}(XY) < 0$.

Proof: From (7), since $\lambda_1, \lambda_2 > 0$, it is sufficient to show that for any $0 < \lambda_3 < 1$,

$$\frac{K(\lambda_1, \lambda_2, \lambda_3)}{\lambda_3 K(\lambda_1 \lambda_3, \lambda_2, \lambda_3)} \times \frac{K(\lambda_1 \lambda_3, \lambda_2 \lambda_3, \lambda_3)}{K(\lambda_1, \lambda_2 \lambda_3, \lambda_3)} > 1. \quad (8)$$

Next, observe that for any $0 < \lambda_3 < 1$, one can write

$$\frac{\lambda_3}{K(\lambda_1, \lambda_2, \lambda_3)} < \frac{1}{K(\lambda_1, \lambda_2, \lambda_3)} < \frac{1}{K(\lambda_1 \lambda_3, \lambda_2, \lambda_3)}. \quad (9)$$

Therefore,

$$\frac{K(\lambda_1, \lambda_2, \lambda_3)}{\lambda_3 K(\lambda_1 \lambda_3, \lambda_2, \lambda_3)} > 1.$$

Similarly, for any $0 < \lambda_3 < 1$,

$$\frac{K(\lambda_1 \lambda_3, \lambda_2 \lambda_3, \lambda_3)}{K(\lambda_1, \lambda_2 \lambda_3, \lambda_3)} > 1. \quad (10)$$

Our result immediately follows on combining (9) and (10). Hence, the proof. Alternatively, it can be verified from the following Corollary (see, Sundt (2000) for details). ■

Corollary 3.1: *The elements of a random vector with infinitely divisible distribution on $\mathbb{N}_m$ are non-negatively correlated, where $\mathbb{N}_m$ denote the set of all $m \times 1$ vectors where all elements are non-negative integers.*

Therefore, X and Y are negative quadrant dependent, i.e. for every pair of decreasing functions $\ell_1(\cdot)$ and $\ell_2(\cdot)$, it follows that $\text{Cov}(\ell_1(X), \ell_2(Y)) \leq 0$. Moreover, in view of the fact that

$$S_{X,Y}(x, y) - S_X(x)S_Y(y) = F_{X,Y}(x, y) - F_X(x)F_Y(y),$$

one may readily observe that $S_{X,Y}(x, y) \leq S_X(x)S_Y(y)$.

Lemma 3.1: *If $(X, Y) \sim \text{BPD}(\lambda_1, \lambda_2, \lambda_3)$, then for any $0 < \lambda_3 \leq 1$, the following holds (for any $(m_1, m_2) \in \mathbb{R}^+$,*

$$\sum_{n=0}^{\infty} \frac{m_1^n}{n!} \exp[m_2 \lambda_3^n] = \sum_{n=0}^{\infty} \frac{m_2^n}{n!} \exp[m_1 \lambda_3^n].$$

Proof: Since the above is an identity, it can be rewritten as (i.e. we need to show equivalently)

$$\sum_{n=0}^{\infty} \frac{m_1^n}{n!} \exp[-m_1 \lambda_3^n] = \sum_{n=0}^{\infty} \frac{m_2^n}{n!} \exp[-m_2 \lambda_3^n].$$

Next, the left-hand side of the above can be written as

$$\sum_{n=0}^{\infty} \frac{m_1^n}{n!} \exp[-m_1 \lambda_3^n] = \sum_{n=0}^{\infty} \frac{m_1^n}{n!} \{\exp[\lambda_3^n]\}^{-m_1} = \exp(m_1),$$

because, for $0 < \lambda_3 \leq 1$, and as n increases and goes to infinity, $\exp[\lambda_3^n] = \exp(0) = 1$. By a similar argument, the right-hand side will be equal to $\exp(m_2)$. Then, on rearrangement of these two terms on both sides of the identity, we get the desired result. Hence, the proof.

Lemma 3.1 will be utilized in establishing the result in Theorem 3.3 which derives the moment generating function (m.g.f.) and the probability generating function (p.g.f.) for the bivariate joint p.m.f. in (1). Notice that similar version of this result has independently observed by [3] [see Exercise problem 4.2(b), page 100]. ■

Theorem 3.2: If $(X, Y) \sim \text{BPD}(\lambda_1, \lambda_2, \lambda_3)$, then the joint factorial moment will be given by

$$E(X_{(r)} Y_{(s)}) = \lambda_1^r \lambda_2^s \lambda_3^{rs} \frac{K(\lambda_1, \lambda_2, \lambda_3)}{K(\lambda_1 \lambda_3^s, \lambda_2 \lambda_3^r, \lambda_3)}.$$

Proof: Simple and thus excluded. ■

Theorem 3.3: If $(X, Y) \sim \text{BPD}(\lambda_1, \lambda_2, \lambda_3)$, then the joint probability generating function (p.g.f.) and the joint moment generating function (m.g.f.) will be

$$G_{X,Y}(s_1, s_2) = \frac{K(\lambda_1, \lambda_2, \lambda_3)}{K_2(\lambda_1 s_1, \lambda_2 s_2, \lambda_3)} = \frac{K(\lambda_1, \lambda_2, \lambda_3)}{K_1(\lambda_1 s_1, \lambda_2 s_2, \lambda_3)},$$

for $|s_1| < 1$, $|s_2| < 1$.

$$M_{X,Y}(t_1, t_2) = \frac{K(\lambda_1, \lambda_2, \lambda_3)}{K_2(\lambda_1 \exp(t_1), \lambda_2 \exp(t_2), \lambda_3)} = \frac{K(\lambda_1, \lambda_2, \lambda_3)}{K_1(\lambda_1 \exp(t_1), \lambda_2 \exp(t_2), \lambda_3)}.$$

Proof: Using Lemma 3.1, our result immediately follows. Therefore, for brevity, the details are excluded. ■

Corollary 3.2: From the bivariate p.g.f. we can obtain the corresponding probabilities by the following formula:

$$P(X = x, Y = y) = \frac{1}{x!y!} \frac{\partial^{x+y}}{\partial s_1^x \partial s_2^y} G_{X,Y}(s_1, s_2) \big|_{s_1=0, s_2=0}.$$

For example,

- Setting $s_1 = 0, s_2 = 0$, $P(X = x, Y = y) = K(\lambda_1, \lambda_2, \lambda_3) K_2^{-1}(0, 0, \lambda_3) = K(\lambda_1, \lambda_2, \lambda_3)$.
- Next, consider

$$P(X = 1, Y = 1) = \frac{\partial^2}{\partial s_1 \partial s_2} G_{X,Y}(s_1, s_2) \big|_{s_1=0, s_2=0}.$$

Now,

$$\frac{\partial^2}{\partial s_1 \partial s_2} G_{X,Y}(s_1, s_2) = (\lambda_1 \lambda_2 \lambda_3) \exp(\lambda_1 \lambda_3 s_1 + \lambda_2 \lambda_3^{t+1} s_2),$$

where $t = x - 1$. Therefore,

$$P(X = 1, Y = 1) = \frac{\partial^2}{\partial s_1 \partial s_2} G_{X,Y}(s_1, s_2) \big|_{s_1=0, s_2=0} = (\lambda_1 \lambda_2 \lambda_3) K(\lambda_1, \lambda_2, \lambda_3)$$

Notice that these individual joint probabilities exactly matches with the joint probabilities that can be obtained from (1) by substituting $(x, y) = \{(0, 0), (1, 1) \dots\}$. Using the joint p.g.f. different moments and product moments can be obtained.

Theorem 3.4: If $(X, Y) \sim \text{BPD}(\lambda_1, \lambda_2, \lambda_3)$, then we have the following results.

- (a) $P(X + Y = n) = K(\lambda_1, \lambda_2, \lambda_3) \sum_{i \geq 0} \binom{n}{i} \lambda_1^i \lambda_2^{n-i} \lambda_3^{i(n-i)}$. In particular $X + Y \sim P(\lambda_1 + \lambda_2)$, if $\lambda_3 = 1$.
- (b)

$$P(X = x | X + Y = n) = \frac{\binom{n}{x} \left(\frac{\lambda_1 \lambda_3^{n-x}}{\lambda_2} \right)^x}{\sum_{i \geq 0} \binom{n}{i} \left(\frac{\lambda_1 \lambda_3^{n-i}}{\lambda_2} \right)^i}.$$

- (c) The regression of X on Y is given by $E(X|Y = y) = \lambda_1 \lambda_3^y$, and The regression of Y on X is given by $E(Y|X = x) = \lambda_2 \lambda_3^x$.

Proof: Part (a) can be obtained from the joint p.g.f. given in Theorem 3.2. Subsequently, part (b) can be obtained on using (a). Proof of part(c) can be obtained immediately by the information that states that both the conditionals are Poisson with respective parameters. Precisely,

- Since, $X|Y = y \sim \text{Poisson}(\lambda_1 \lambda_3^y)$, therefore, the regression of X on Y will be obtained as $X = E(X|Y = y) = \lambda_1 \lambda_3^y$.
- Similarly, since, $Y|X = x \sim \text{Poisson}(\lambda_2 \lambda_3^x)$, therefore, the regression of Y on X will be obtained as $Y = E(Y|X = x) = \lambda_2 \lambda_3^x$.

■

Theorem 3.5: If $(X, Y) \sim \text{BPD}(\lambda_1, \lambda_2, \lambda_3)$, then we have the following stochastic ordering results of the family of univariate Poisson distribution arising from the bivariate distribution in (1).

- (a) If $\lambda_1 > \lambda_2$, then for any $0 < \lambda_3 < 1$, X is stochastically larger than Y . This also implies that under this parametric restriction, Y is smaller than X in the hazard rate order, mean residual life order, and the likelihood ratio order.
- (b) If $\lambda_1 < \lambda_2$, then for any $0 < \lambda_3 < 1$, Y is stochastically larger than X . This also implies that under this parametric restriction, X is smaller than Y in the hazard rate order, mean residual life order, and the likelihood ratio order.
- (c) If $\lambda_1 < \lambda_2$, then for any $0 < \lambda_3 < \min\{\lambda_1, \lambda_2\} < 1$, Y is stochastically larger than X . This also implies that under this parametric restriction, X is smaller than Y in the hazard rate order, mean residual life order, and the likelihood ratio order.
- (d) If $\lambda_1 > \lambda_2$, then for any $1 > \lambda_3 > \max\{\lambda_1, \lambda_2\}$, X is stochastically larger than Y . This also implies that under this parametric restriction, Y is smaller than X in the hazard rate order, mean residual life order, and the likelihood ratio order.

Proof: The proof is quite simple and hence, the details avoided.

■

Theorem 3.6: If $(X, Y) \sim \text{BPD}(\lambda_1, \lambda_2, \lambda_3)$, then the marginals of X and Y exhibit over-dispersion.

Proof: For over-dispersion, we need to show that the variance is greater than the mean. We start with the mean and variance of X . The case of Y can be similarly dealt. Regarding the mean and variance of X , the condition of over-dispersion reduces to establish that

$$\lambda_1 K(\lambda_1, \lambda_2, \lambda_3) \left[\frac{1}{K_1} + \lambda_1 \left\{ \frac{1}{K(\lambda_1, \sqrt{\lambda_2} \lambda_3, \lambda_3)} - \frac{K(\lambda_1, \lambda_2, \lambda_3)}{K_1^2} \right\} \right] > \frac{\lambda_1 K(\lambda_1, \lambda_2, \lambda_3)}{K_1},$$

which is, equivalently can be written as

$$\frac{1}{K(\lambda_1, \sqrt{\lambda_2} \lambda_3, \lambda_3)} > \frac{K(\lambda_1, \lambda_2, \lambda_3)}{K_1^2}. \quad (11)$$

Next, note that (11) is true since $\text{Var}(X) \geq 0$. Hence, the proof. ■

Theorem 3.7: $(X, Y) \sim \text{BPD}(\lambda_1, \lambda_2, \lambda_3)$, belongs to three parameter exponential family.

Proof: The joint p.m.f. of $(X, Y) \sim \text{BPD}(\lambda_1, \lambda_2, \lambda_3)$, will be a member of the 3-parameter exponential family if its p.m.f. can be expressed in the form

$$P(X = x, Y = y) = (x!y!)^{-1} \exp \left[\sum_{j=1}^3 \eta_j(\lambda_1, \lambda_2, \lambda_3) T_j(x, y) - A(\lambda_1, \lambda_2, \lambda_3) \right]. \quad (12)$$

From (13), it is easy to observe that the proposed distribution belongs to the exponential family by rewriting the p.m.f. as

$$P(X = x, Y = y) = [h(x, y)] \times \exp [x \log \lambda_1 + y \log \lambda_2 + xy \log \lambda_3 - \log K^{-1}(\lambda_1, \lambda_2, \lambda_3)], \quad (13)$$

and identifying

$$\begin{aligned} h(x, y) &= (x!y!)^{-1}, \\ \eta_1(\lambda_1, \lambda_2, \lambda_3) &= \lambda_1, \eta_2(\lambda_1, \lambda_2, \lambda_3) = \lambda_2, \eta_3(\lambda_1, \lambda_2, \lambda_3) = \lambda_3, \\ T_1(x, y) &= x, T_2(x, y) = y, T_3(x, y) = xy, \\ \text{and } A(\lambda_1, \lambda_2, \lambda_3) &= \log K^{-1}(\lambda_1, \lambda_2, \lambda_3). \end{aligned}$$

Thus, based on a sample of size m from BPD, $(\sum x, \sum y, \sum xy)$ is completely sufficient for $(\lambda_1, \lambda_2, \lambda_3)$. ■

Distributions belonging to exponential family enjoy many properties. For example mean, variance, co-variance and moment generating functions can be easily derived using differentiation's of $A(\lambda_1, \lambda_2, \lambda_3)$. Moreover using Lehmann Scheffe result, it may be possible to derive UMVUE of the parameters, provided we get hold of function of T that is unbiased for the parameter. Even otherwise one can derive MVUE implementing bias correction.

Theorem 3.8: If $(X, Y) \sim \text{BPD}(\lambda_1, \lambda_2, \lambda_3)$, then the modal value of the distribution will be as follows

- For strictly integer-valued parameters, the modal value will be at $(\lambda_1 + \lambda_3 - 1, \lambda_2 + \lambda_3 - 1)$.
- For any set of parameters that contain non-integers, the modal value will be at $(\lfloor * \rfloor \lambda_1 + \lambda_3 - \eta_1, \lfloor * \rfloor \lambda_2 + \lambda_3 - \eta_2)$, where $\lfloor * \rfloor x$ is the greatest integer function, and $\eta_i = 0, 1$ is an indicator function.

Proof: Straightforward and hence omitted. ■

Theorem 3.9: If $(X, Y) \sim \text{BPD}(\lambda_1, \lambda_2, \lambda_3)$, then

$$P(X < Y) = \lambda_2 [K(\lambda_1, \lambda_2, \lambda_3)]^2 \left(\sum_{j=0}^{\infty} \sum_{x=0}^j \left(\frac{\exp(\lambda_1 \lambda_3^{j+1})}{j!} \right) \left(\frac{\exp(\lambda_2 \lambda_3^x)}{x!} \right) \right).$$

Proof: Observe that

$$\begin{aligned} P(X < Y) &= \sum_{j=0}^{\infty} P(Y = j+1) P(X \leq j) \\ &= \sum_{j=0}^{\infty} \left(K(\lambda_1, \lambda_2, \lambda_3) \left(\frac{\lambda_2^{j+1} \exp(\lambda_1 \lambda_3^{j+1})}{j!} \right) \right) \\ &\quad \times \left(\sum_{x=0}^j \left(K(\lambda_1, \lambda_2, \lambda_3) \frac{\lambda_1^x \exp(\lambda_2 \lambda_3^x)}{x!} \right) \right) \\ &= \lambda_2 [K(\lambda_1, \lambda_2, \lambda_3)]^2 \left(\sum_{j=0}^{\infty} \sum_{x=0}^j \left(\frac{\lambda_2^j \exp(\lambda_1 \lambda_3^{j+1})}{j!} \right) \left(\frac{\lambda_1^x \exp(\lambda_2 \lambda_3^x)}{x!} \right) \right). \end{aligned} \tag{14}$$

Note: Observe that since $0 < \lambda_3 \leq 1$, one can obtain an upper bound inequality of $P(X < Y)$ by setting $\lambda_3 = 1$, which will be as follows:

$$P(X < Y) \leq \lambda_2 \exp(3\lambda_1 + 2\lambda_2) \left(\sum_{j=0}^{\infty} \frac{\lambda_2^j}{j!} \sum_{x=0}^j \frac{\exp(-\lambda_1) \lambda_1^x}{x!} \right),$$

on using the fact that $K(\lambda_1, \lambda_2, 1) = \exp(\lambda_1 + \lambda_2)$. Furthermore, on using the inequality result related to a Poisson distribution in Sort [23], we may additionally write the above

inequality as

$$\begin{aligned}
 P(X < Y) &\leq \lambda_2 \exp(3\lambda_1 + 2\lambda_2) \left(\sum_{j=0}^{\infty} \frac{\lambda_2^j}{j!} \sum_{x=0}^j \frac{\exp(-\lambda_1)\lambda_1^x}{x!} \right) \\
 &> \lambda_2 \exp(3\lambda_1 + 2\lambda_2) \left(\sum_{j=0}^{\infty} \frac{\lambda_2^j}{j!} \left[1 - \frac{\exp(-H(\lambda_1, j))}{\max\{2, \sqrt{4\pi H(\lambda_1, j)}\}} \right] \right),
 \end{aligned}$$

where $H(\lambda_1, j)$ is the KL divergence between two Poisson distributed random variables with respective means λ_1 and j . $R = P(X < Y)$ is known as the stress-strength reliability in engineering where the random variables X and Y , respectively, represent the stress and strength associated with a system. This measure is also useful in a probabilistic assessment of inequality in two phenomena X and Y . While in most cases X and Y are assumed to be independent in real life there may be dependence between X and Y . Stress-strength reliability with both variables having independent Poisson distribution was discussed in [4]. As such, the result of Theorem 9 has the potential to be used in such contexts.

4. Statistical inference

4.1. Maximum likelihood estimation

In this section, we consider the maximum-likelihood estimation of the unknown parameters of a $\text{BPD}(\lambda_1, \lambda_2, \lambda_3)$ model based on a random sample of size m , namely $\mathbb{C} = (x_1, y_1), \dots, (x_m, y_m)$. The proposed bivariate Poisson model has three parameters. It is observed that the MLEs of the unknown parameters can be obtained by solving a three-dimensional optimization problem. However, we have used Mathematica to evaluate the estimates and the corresponding performance of the MLE in this case. For notational simplicity, we will indicate the parameter vector by $\Delta = (\lambda_1, \lambda_2, \lambda_3)$. Based on the random sample $\mathbb{C}$ as mentioned above and using (2), the log-likelihood function can be written as

$$\ell(\Delta|\mathbb{C}) = mK(\lambda_1, \lambda_2, \lambda_3) + \log \lambda_1 \sum_{i=1}^m x_i + \log \lambda_2 \sum_{i=1}^m y_i + \log \lambda_3 \sum_{i=1}^m x_i y_i. \quad (15)$$

The MLEs of the unknown parameters can be obtained by maximizing (15) with respect to Δ . It requires solving the following three non-linear equations

$$\frac{\partial \ell(\Delta|\mathbb{C})}{\partial \lambda_1} = 0, \quad \frac{\partial \ell(\Delta|\mathbb{C})}{\partial \lambda_2} = 0, \quad \frac{\partial \ell(\Delta|\mathbb{C})}{\partial \lambda_3} = 0. \quad (16)$$

Clearly, they cannot be obtained in explicit forms. Newton-Raphson method may be used to solve these non-linear equations. One may consider an appropriate EM algorithm in this case as an alternative. However, as described later on, we will consider a copula-based estimation and simulation for a $\text{BPD}(\lambda_1, \lambda_2, \lambda_3)$ for varying parameter choices.

4.2. Likelihood ratio test for independence of the random variables

As suggested earlier, if $\lambda_3 = 1$, then X and Y are independent. Thus, we may conduct the test of independence of the two random variables X and Y by testing the null hypothesis

$$H_0 : \lambda_3 = 1 \quad \text{vs} \quad H_a : \lambda_3 \neq 1.$$

The likelihood ratio statistic is given by

$$\Lambda = \frac{\max_{\Delta \in \Omega_0} L(\Delta | \mathbb{C})}{\max_{\Delta \in \Omega} L(\Delta | \mathbb{C})},$$

where $L(\cdot)$ stands for the likelihood function, Ω_0 is the space of parameters under H_0 , and Ω is the space of parameters with no restrictions. It is well known that for large sample sizes, we have

$$-2 \log \Lambda \stackrel{asympt}{\sim} \chi_{r-r_0}^2,$$

where, in our testing procedure, $r = 3$ is the number of free parameters under the alternative hypothesis and $r_0 = 2$ is the number of free parameters under the null hypothesis. Using the methodologies describe subsection 4.1, it is possible to implement this test and subsequently test the independence of the random variables X and Y . In a similar way, one can develop of testing the following hypothesis to check for symmetry (see item 3 on section 2)

$$H_0 : \lambda_1 = \lambda_2 \quad \text{vs} \quad H_a : \lambda_1 \neq \lambda_2.$$

4.3. Copula-based simulations

In order to assess the importance and efficacy of the proposed distribution, we consider a copula-based simulation study. We generate samples of size 10000 from two copulas which allow to define a negative correlation, more precisely (i) a Bivariate Normal (BN) copula with correlation ρ , and (ii) a Farlie–Gumbel–Morgenstern (FGM) copula with the dependence parameter α ; both combined with two marginal Poisson distributions with parameters m_1 and m_2 , respectively. The purpose here is to understand how the bivariate distribution described in (1) may be used to model bivariate data arising from a BN or FGM copulas with Poisson marginals.

The simulation from a copula-based distribution is quite straightforward when the marginals are continuous. However, when the marginals are discrete or a mixture of discrete and continuous, a more careful approach is required, for reference, see [11,25]. In this section, we use the BN and the FGM copulas combined with marginal Poisson distributions with parameters m_1 and m_2 and corresponding cumulative distribution functions (c.d.f.s) given by $F_1(x_1)$ and $F_2(x_2)$. This approach generates the following joint distribution:

$$F(x_1, x_2) = C_\theta(u_1 = F_1(x_1), u_2 = F_2(x_2)),$$

where C_θ stands for the copula considered (BN or FGM) with the association parameter θ . To generate sample from this distribution, we have to first sample the vector $\underline{u} = (u_1, u_2)$ and then, using the inverses $F_1^{-1}(\cdot)$ and $F_2^{-1}(\cdot)$, obtain $\underline{x} = (x_1, x_2) = (F_1^{-1}(u_1), F_2^{-1}(u_2))$.

Table 1. Analysis of the fit of the BPD distribution to the data generated from a Binormal copula, with correlation ρ , of two Poisson distributions with parameters $m_1 = 2$ and $m_2 = 3$.

ρ	$\{\hat{\lambda}_1, \hat{\lambda}_2, \hat{\lambda}_3\}$	Correlation		p -value
		BPD	Samp.	
−0.9	{3.97, 4.75, 0.76}	−0.54	−0.82	≈ 0.00
−0.8	{3.81, 4.61, 0.78}	−0.51	−0.72	≈ 0.00
−0.7	{3.63, 4.46, 0.79}	−0.48	−0.64	≈ 0.00
−0.6	{3.51, 4.36, 0.81}	−0.45	−0.58	≈ 0.00
−0.5	{3.22, 4.14, 0.84}	−0.39	−0.46	≈ 0.00
−0.4	{3.04, 3.96, 0.86}	−0.34	−0.38	≈ 0.00
−0.3	{2.76, 3.69, 0.89}	−0.26	−0.28	≈ 0.00
−0.2	{2.47, 3.49, 0.93}	−0.18	−0.19	0.21
−0.1	{2.24, 3.25, 0.96}	−0.10	−0.10	0.76
0.1	{2.03, 3.02, 1.00}	0.00	0.10	≈ 0.00
0.2	{4.94, 10.13, 1.0}	0.00	0.19	≈ 0.00

In Appendix A.1.4 of [26], the authors present an example of a method that may be used to simulate discrete dependent Poisson variables using a specific copula. For illustrative purposes, we have provided a scenario of simulating discrete dependent Poisson variables with means m_1 and m_2 from a bivariate Normal copula in the Appendix. For further details on discrete copulas characterization and simulation see [18,19,25] and the references cited therein. An interesting review of multivariate distributions for count data derived from the Poisson distribution can be found in [11]. For the simulations here, Mathematica software is used to implement the methodology.

In our simulation study, for the BN copula, the correlation parameter ρ varied between -0.9 to 0.2 , and for the FGM copula, the parameter α varied also from -0.9 to 0.2 . The MLEs for λ_1 , λ_2 and λ_3 were obtained using the procedure described in Subsection 4.1, using the expression in (13). To test the fit of the BPD distribution to the data generated from the BN and FGM copulas, we consider the χ^2 goodness-of-fit test for bivariate discrete distributions following the suggestions in [16], namely we considered the ordered expected-frequencies procedure described in Section 3 of the same reference. From Tables 1 to 4, we may observe that the BPD distribution may be a useful tool to model bivariate data, which may be characterized by a BN or FGM copula with a small negative correlation in the range $(-0.2, 0)$. When the negative correlation is weaker, (smaller than -0.2) the null hypothesis, which states that the population distribution is BPD, is rejected.

5. Real data applications

5.1. English premier football league data

Here, we consider data from the English Premier Football League [available in the web page – <https://datahub.io/sports-data/english-premier-league#data> –] for five(5) seasons, 2014/15, 2015/16, 2016/17, 2017/18 and 2018/19. Each season giving us 380 pairs of counts, namely the goals scored in each game in the form (*Full Time Home Team Goal, Full time Away Team Goal*) with negative correlations. We conjecture that the BPD distribution proposed in this paper will provide a reasonably good fit to model this data. In Table 5, we present a descriptive summary of the data from each of the five (5) seasons, and provide

Table 2. Analysis of the fit of the BPD distribution to the data generated from a Binormal copula, with correlation ρ , of two Poisson distributions with parameters $m_1 = 5$ and $m_2 = 10$.

ρ	$\{\hat{\lambda}_1, \hat{\lambda}_2, \hat{\lambda}_3\}$	Correlation		p -value
		BPD	Samp.	
−0.9	{11.3, 15.04, 0.92}	−0.57	−0.87	≈ 0.00
−0.8	{10.8, 14.69, 0.92}	−0.54	−0.78	≈ 0.00
−0.7	{10.1, 14.15, 0.93}	−0.49	−0.67	≈ 0.00
−0.6	{9.63, 13.85, 0.93}	−0.46	−0.58	≈ 0.00
−0.5	{8.91, 13.36, 0.94}	−0.41	−0.49	≈ 0.00
−0.4	{8.12, 12.77, 0.95}	−0.35	−0.40	≈ 0.00
−0.3	{7.38, 12.16, 0.96}	−0.28	−0.30	≈ 0.00
−0.2	{6.71, 11.57, 0.97}	−0.21	−0.21	0.44
−0.1	{5.82, 10.80, 0.98}	−0.11	−0.11	0.97
0.1	{5.00, 9.96, 1.00}	0.00	0.09	≈ 0.00
0.2	{5.01, 9.97, 1.00}	0.00	0.20	≈ 0.00

Table 3. Analysis of the fit of the BPD distribution to the data generated from a FGM copula, with correlation ρ , of two Poisson distributions with parameters $m_1 = 2$ and $m_2 = 3$.

ρ	$\{\hat{\lambda}_1, \hat{\lambda}_2, \hat{\lambda}_3\}$	Correlation		p -value
		BPD	Samp.	
−0.9	{2.69, 3.68, 0.90}	−0.25	−0.26	≈ 0.00
−0.8	{2.65, 3.59, 0.91}	−0.23	−0.24	≈ 0.00
−0.7	{2.57, 3.52, 0.92}	−0.20	−0.20	0.02
−0.6	{2.43, 3.44, 0.93}	−0.17	−0.17	≈ 0.00
−0.5	{2.36, 3.34, 0.95}	−0.13	−0.14	0.07
−0.4	{2.31, 3.30, 0.95}	−0.12	−0.12	0.16
−0.3	{2.23, 3.19, 0.96}	−0.09	−0.09	0.05
−0.2	{2.16, 3.15, 0.97}	−0.06	−0.06	0.65
−0.1	{2.09, 3.10, 0.99}	−0.03	−0.03	0.58
0.1	{2.02, 3.01, 1.00}	0.00	0.05	≈ 0.00
0.2	{2.00, 3.02, 1.00}	0.00	0.07	≈ 0.00

Table 4. Analysis of the fit of the BPD distribution to the data generated from a FGM copula, with correlation ρ , of two Poisson distributions with parameters $m_1 = 5$ and $m_2 = 10$.

ρ	$\{\hat{\lambda}_1, \hat{\lambda}_2, \hat{\lambda}_3\}$	Correlation		p -value
		BPD	Samp.	
−0.9	{7.31, 12.10, 0.96}	−0.26	−0.29	≈ 0.00
−0.8	{7.18, 11.93, 0.96}	−0.25	−0.27	≈ 0.00
−0.7	{6.73, 11.60, 0.97}	−0.21	−0.23	≈ 0.00
−0.6	{6.58, 11.48, 0.97}	−0.19	−0.20	0.02
−0.5	{6.06, 11.03, 0.98}	−0.14	−0.14	0.02
−0.4	{6.02, 11.01, 0.98}	−0.13	−0.14	0.38
−0.3	{5.64, 10.58, 0.99}	−0.09	−0.09	0.26
−0.2	{5.58, 10.52, 0.99}	−0.08	−0.08	0.34
−0.1	{5.38, 10.35, 0.99}	−0.05	−0.05	0.48
0.1	{5.01, 9.93, 1.00}	0.00	0.04	0.13
0.2	{4.98, 9.96, 1.00}	0.00	0.06	0.12

plots of the observed histograms for these bivariate data in Figure 7 for each of the 5 seasons, respectively. As expected, pairwise negative correlations can be observed for each of the 5 seasons.

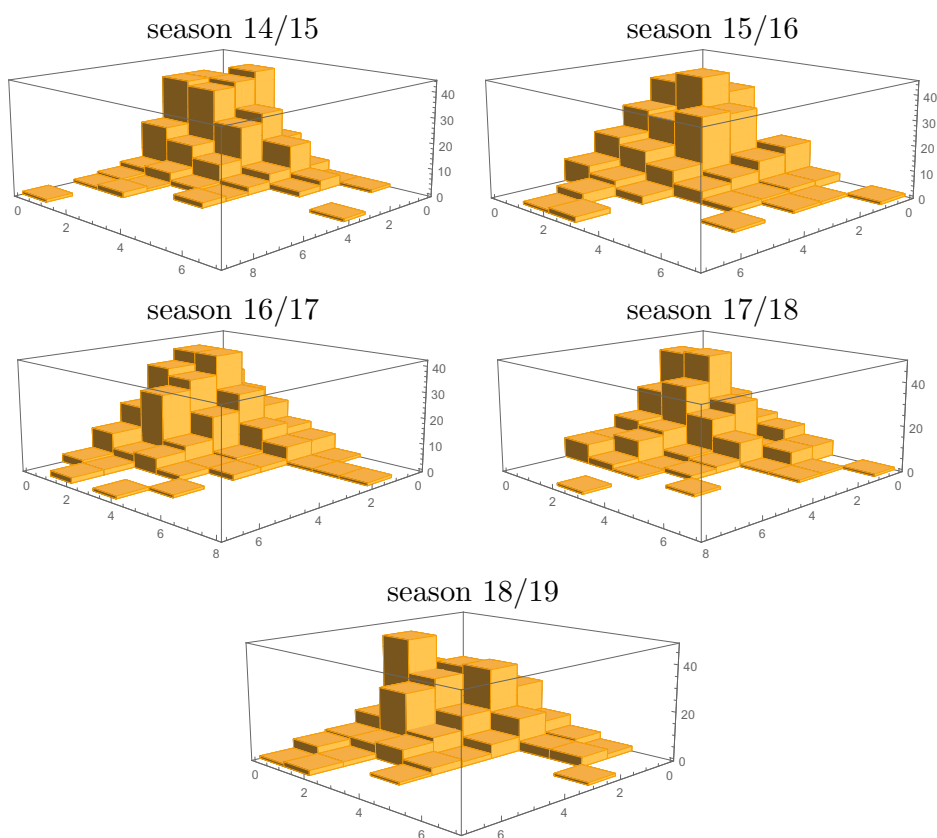


Figure 7. Histograms for the bivariate data of goals scored, (*Home team*, *Away team*), for the seasons 2014/15, 2015/16, 2016/17, 2017/18 and 2018/19 for the English Premier Football League.

The results from the data fitting of the BPD distribution to the data for the seasons 2014/15, 2015/16, 2016/17, 2017/18 and 2018/19 is presented in the Table 6 reveal that, with the exception of one season namely, 2017/18, the BPD distribution provides an adequate fit to the bivariate pair data arising from the rest of the for seasons.

5.2. Data of counts of surface and interior faults in 100 lenses

Aitchison and Ho [2] have analyzed a bivariate count data (Table I from [2]) that originated as counts of surface faults (X) and interior faults (Y) of lenses. The same data (Table VI from [15]) was also studied by [15]. In Table 2 of [2] and Table VII of [15], the both the authors considered five different bivariate distributions to model data of counts of surface and interior faults in 100 lenses. The results are compared based on the log-likelihood, AIC, and BIC criteria. In this subsection, we have re-analyzed this dataset since it is negatively correlated, more precisely $\rho = -0.175$. In Figure 8, we present the histogram of the data, and in Table 7, we present the observed values of the log-likelihood, AIC and BIC obtained when this data set is fitted with a BPD distribution proposed in this paper. Comparing our findings with those in [2] and [15], it is observed that the BPD distribution in (1) presents

Table 5. Descriptive analysis of the data of goals scored, (*Home team, Away team*), for the seasons 2014/15, 2015/16, 2016/17, 2017/18 and 2018/19 of the English Premier Football League.

Seasons		Min	$q_{0.25}$	Median= $q_{0.5}$	Mean	$q_{0.75}$	Max	Correlation
2014/15	Home team	0	1.0	1.0	1.5	2.0	8.0	−0.030
	Away team	0	0	1.0	1.1	2.0	6.0	
2015/16	Home team	0	1.0	1.0	1.5	2.0	6.0	−0.022
	Away team	0	0	1.0	1.2	2.0	6.0	
2016/17	Home team	0	1.0	1.0	1.6	2.0	6.0	−0.132
	Away team	0	0	1.0	1.2	2.0	7.0	
2017/18	Home team	0	1.0	1.0	1.5	2.0	7.0	−0.130
	Away team	0	0	1.0	1.1	2.0	6.0	
2018/19	Home team	0	1.0	1.0	1.6	2.0	6.0	−0.178
	Away team	0	0	1.0	1.3	2.0	6.0	

Table 6. Analysis of the fit of the BPD distribution to the data of goals scored, (*Home team, Away team*), for the seasons 2014/15, 2015/16, 2016/17, 2017/18 and 2018/19 for the English Premier Football League.

Season	$\{\hat{\lambda}_1, \hat{\lambda}_2, \hat{\lambda}_3\}$	Correlation		p -value
		BPD	data	
2014/15	{1.52, 1.13, 0.97}	−0.03	−0.03	0.36
2015/16	{1.52, 1.24, 0.99}	−0.02	−0.02	0.05
2016/17	{1.82, 1.43, 0.89}	−0.15	−0.13	0.27
2017/18	{1.75, 1.37, 0.89}	−0.15	−0.13	0.0009
2018/19	{1.86, 1.55, 0.86}	−0.19	−0.18	0.10

Table 7. AIC and BIC values for BPD distribution fit of the data of counts of surface and interior faults in 100 lenses in [2, 15].

$\{\hat{\lambda}_1, \hat{\lambda}_2, \hat{\lambda}_3\}$	$\log$ -likelihood	AIC	BIC
{2.69, 3.68, 0.90}	−445.0	896.0	903.8

better results, in terms of the AIC and BIC measures than McKendrick–Wicksell bivariate Poisson distribution [2], Bivariate Poisson distribution proposed by Holgate [10] and bivariate geometric [15]. It is important here to note that the other distributions compared in the said papers which apparently provide a better fit than the BPD all have four or more parameters and hence more parsimony as expected.

5.3. Seeds and plants grown data

As a last example of application, we consider the data about seeds and plants grown in [22] and also in Table 1 in [14]. The data set reports the number of seeds and plants grown over a plot of size five square feet. The sample correlation is negative and equal -0.094 , thus this is the kind of scenario in which the proposed BPD distribution may be useful to model the data. In Figure 9 we present the histogram of the data and in Table 8, we present

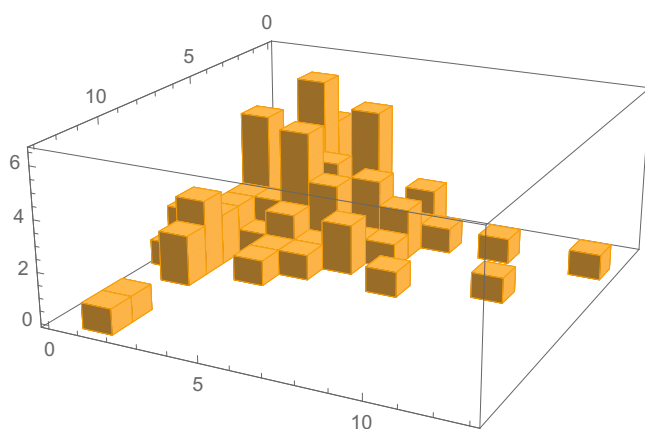


Figure 8. Data of counts of surface and interior faults in 100 lenses.

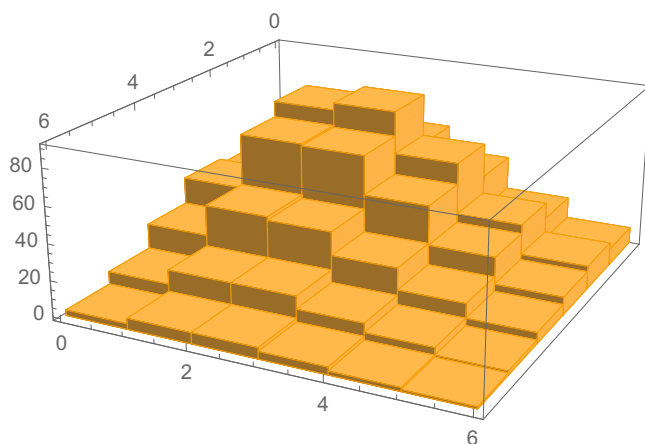


Figure 9. Histogram of the data of seeds and plants grown.

Table 8. Analysis of the fit of the BPD distribution to the data of seeds and plants grown.

$\{\hat{\lambda}_1, \hat{\lambda}_2, \hat{\lambda}_3\}$	Correlation		p -value
	BPD	data	
$\{1.85, 2.17, 0.96\}$	−0.08	−0.09	0.97

the estimated values of λ_1 , λ_2 and λ_3 , together with the correlation measures and with the p -value of χ^2 goodness-of-fit test for the BPD distribution.

One may observe that the sample and population correlations are similar and that the p -value of the χ^2 goodness-of-fit test for the BPD distribution is close to 1, indicating an excellent fit of the BPD distribution to the data. In [14], the authors also perform the goodness-of-fit test for a bivariate Poisson distribution and also in the case of independence. However, since in the bivariate case, there are different ways of grouping the classes we will not compare our results with the ones in [14].

6. Conclusion

The statistical analysis of bivariate count data has proved to be challenging, because of the lack of a parametric class of distributions supporting a rich-enough correlation structure. Application of bivariate Poisson distribution as a tool to model bivariate count data is not new in the literature (see [2] and the references cited there in), but, several such earlier proposed models has the drawback of its inability to model data sets exhibiting negative correlation. For example, the bivariate Poisson distribution given by [5] and many other authors later on exhibit only a positive correlation by the very nature of the genesis of the distribution. However, in the present paper, we have considered a special type of bivariate Poisson distribution described by [3] in which the construction of a bivariate distribution starts with two given conditionals which belongs to the same family (in our case Poisson) with appropriate dependence parameters. We have established that under certain conditions of the parameters, this model will exhibit negative correlation, in fact always as suggested by [3]. That is one major objective for this study among others. We have also fitted the BPD distribution to seven data sets exhibiting negative correlation with a satisfactory level of performance as can be verified by the associated p -values. The other striking feature of this BPD distribution in (1) is that it is free from the limitation that the derivation of any structural properties including the derivation of marginal p.m.f.s that involves evaluation of incomplete gamma function unlike other bivariate Poisson distributions that exist in the literature. It is important to see how the model in (1) can be generalized to the multivariate case. A substantive amount of work is required in this direction to see the applicability of such models in practice. However, extension to the multivariate version of this discrete distribution should be motivated from a real-world perspective.

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Appendix

Here, we provide for illustrative purposes, as to how we can simulate random samples from a bivariate discrete distribution via a bivariate dependent copula, namely from a bivariate normal (Gaussian) copula. The bivariate normal (or Gaussian) copula with dependence parameter ρ is defined by (for

details, see Section 2.3 of [17]):

$$C(u, v; \rho) = \Phi_2(\Phi^{-1}(u), \Phi^{-1}(v); \rho), \quad (\text{A1})$$

where

$$\phi_2(x, y; \rho) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{x^2 + y^2 - 2\rho xy}{2(1-\rho^2)}\right), \quad \Phi_2(h, k; \rho) = \int_{-\infty}^h \int_{-\infty}^k \phi_2(x, y; \rho) \, dy \, dx,$$

are the density and distribution function of the bivariate standard normal distribution with correlation parameter $\rho \in [-1, 1]$. We focus on simulating discrete Poisson variables using a method based on Devroye's technique of sequential search [7]. The algorithm is follows:

- First, we draw correlated uniform random variables (u, v) from the BVN copula in (A1) using a well known standard procedure given as follows:
 1. Generate two independently distributed $N(0, 1)$ variables, say u and v .
 2. Set $y_1 = u$, and $y_2 = u \times \rho + v\sqrt{1 - \rho^2}$.
 3. Set $u = \Phi(y_1)$, and $v = \Phi(y_2)$, where Φ is the cumulative distribution function of the standard normal distribution. Then, the pair (u, v) are uniformly distributed variables drawn from the Gaussian copula $C(u, v; \rho)$.
- Next, set the Poisson mean m_1 such that $P(Y_1 = 0) = \exp(-m_1)$.
- Set $Y_1 = 0$, $A_0 = \exp(-m_1)$, and $W = A_0$.
- If $u < W$, then Y_1 remains equal to 0.
- If $u > W$, then proceed sequentially as follows: While $u > W$, replace—(a) $Y_1 \leftarrow Y_1 + 1$, (b) $A_0 \leftarrow m_1 \frac{A_0}{Y_1}$, (c) $W \leftarrow W + A_0$. Continue this process until $u < W$.

These steps produce a simulated variable Y_1 with Poisson distribution with mean m_1 . To obtain draws of the second Poisson variable Y_2 , (say) replace u and m_1 with v and m_2 and repeat the steps above. Consequently, the pair (Y_1, Y_2) are jointly distributed Poisson variables with means m_1 and m_2 .